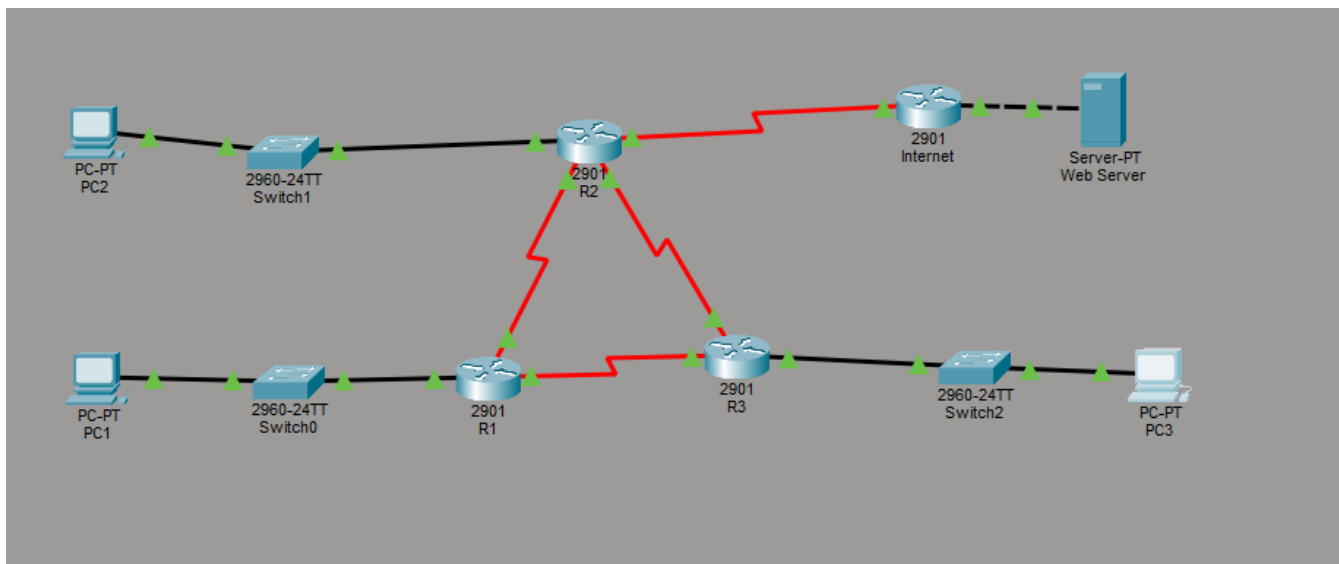


lab10 2.4.11 Packet Tracer - Modify Single-Area OSPFv2 – ILM

นายจิรภัทร แก้วดี 673380577-9 Sec 2

ขออนุญาตทำเป็นแบบจำลองทั้งหมดเนื่องจากคาบเรียนตรงกับวิชาGe ทำให้ไม่สามารถใช้อุปกรณ์จริงได้



Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	G0/0	172.16.1.1	255.255.255.0	N/A
	S0/0/0	172.16.3.1	255.255.255.252	
	S0/0/1	192.168.10.5	255.255.255.252	
R2	G0/0	172.16.2.1	255.255.255.0	N/A
	S0/0/0	172.16.3.2	255.255.255.252	
	S0/0/1	192.168.10.9	255.255.255.252	
	S0/1/0	209.165.200.225	255.255.255.224	
R3	G0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	192.168.10.6	255.255.255.252	
	S0/0/1	192.168.10.10	255.255.255.252	
PC1	NIC	172.16.1.2	255.255.255.0	172.16.1.1
PC2	NIC	172.16.2.2	255.255.255.0	172.16.2.1
PC3	NIC	192.168.1.2	255.255.255.0	192.168.1.1
Web Server	NIC	64.100.1.2	255.255.255.0	64.100.1.1

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interf g0/0
R1(config-if)#ip address 172.16.1.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#exit
R1(config)#interface s0/0/0
R1(config-if)#ip address 172.16.3.1 255.255.255.252
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R1(config-if)#exit
R1(config)#interface s0/0/1
R1(config-if)#
R1(config-if)#ip address 192.168.10.5 255.255.255.252
R1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
R1(config-if)#
```

เข้าสู่ Global Configuration Mode ตั้งชื่อ hostname เป็น R1 จากนั้นกำหนด IP address ให้กับอินเทอร์เฟซ G0/0 (172.16.1.1/24), S0/0/0 (172.16.3.1/30) และ S0/0/1 (192.168.10.5/30) พร้อมเปิดใช้งานอินเทอร์เฟซด้วยคำสั่ง no shutdown โดย G0/0 ขึ้นสถานะ up ทันที ส่วน Serial ทั้งสองยังเป็น down เนื่องจากอีกฝั่งของลิงก์ยังไม่ได้ตั้งค่า

```

Router>enable
Router#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface g0/0
R2(config-if)#ip address 172.16.2.1 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R2(config-if)#exit
R2(config)#interface s0/0/0
R2(config-if)#ip address 172.16.3.2 255.255.255.252
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

R2(config-if)#exit
R2(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R2(config)#interface s0/0/1
R2(config-if)#ip address 209.165.200.225 255.255.255.224
R2(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
R2(config-if)#exit
R2(config)#

```

ตั้ง hostname เป็น R2 และกำหนด IP ให้ G0/0 (172.16.2.1/24), S0/0/0 (172.16.3.2/30) และ S0/0/1 (209.165.200.225/30) จะเห็นว่า S0/0/0 ขึ้นสถานะ up/up ทันทีเนื่องจาก R1 ได้ตั้งค่าฝั่งตรงข้ามไว้แล้ว ยืนยันว่าลิงก์ R1-R2 เชื่อมต่อสำเร็จ

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R3
R3(config)#interface g0/0
R3(config-if)#ip address 192.168.1.1 255.255.255.0
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R3(config-if)#exit
R3(config)#interface s0/0/0
R3(config-if)#ip address 192.168.10.6 255.255.255.252
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

R3(config-if)#exit
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R3(config)#interface s0/0/1
R3(config-if)#ip address 192.168.10.10 255.255.255.252
R3(config-if)#no shutdown

R3(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to up

R3(config-if)#exit
R3(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/1, changed state to up

```

ตั้ง hostname เป็น R3 และกำหนด IP ให้ G0/0 (192.168.1.1/24), S0/0/0 (192.168.10.6/30) และ S0/0/1 (192.168.10.10/30) ทุกอินเทอร์เฟซขึ้นสถานะ up/up ทันที เนื่องจากฝั่งตรงข้าม (R1 และ R2) ได้ตั้งค่าไว้เรียบร้อยแล้ว

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Internet
Internet(config)#interface s0/0/0
Internet(config-if)#ip address 209.165.200.226 255.255.255.224
Internet(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
Internet(config-if)#exit
Internet(config)#interface g0/0
Internet(config-if)#ip address 64.100.1.1 255.255.255.0
Internet(config-if)#no shutdown

Internet(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Internet(config-if)#exit
Internet(config)#
```

ตั้ง hostname เป็น Internet และกำหนด IP ให้ S0/0/0 (209.165.200.226/27 เชื่อมกับ R2) และ G0/0 (64.100.1.1/24 เชื่อมกับ Web Server) เพื่อจำลองเส้นทางออกอินเทอร์เน็ตไปยัง Web Server ภายนอก

Objectives

Part 1: Modify OSPF Default Settings

Part 2: Verify Connectivity

Scenario

In this activity, OSPF is already configured and all end devices currently have full connectivity. You will modify the default OSPF routing configurations by changing the hello and dead timers and adjusting the bandwidth of a link. Then you will verify that full connectivity is restored for all end devices.

Instructions

Part 1: Modify OSPF Default Settings

```
R1(config-if)#router ospf 1
R1(config-router)#network 172.16.1.0 0.0.0.255 area 0
R1(config-router)#network 172.16.3.0 0.0.0.3 area 0
R1(config-router)#network 192.168.10.4 0.0.0.3 area 0
R1(config-router)#
```

Copy

Paste

เปิดใช้งาน OSPF process 1 และประกาศ network statement ครอบคลุมทุก subnet ที่เชื่อมต่อกับ R1 (172.16.1.0/24, 172.16.3.0/30, 192.168.10.4/30) เข้าสู่ area 0 เพื่อให้ R1 แลกเปลี่ยนเส้นทางกับ router อื่นในเครือข่าย

```

R2(config)#router ospf 1
R2(config-router)#network 172.16.2.0 0.0.0.255 area 0
R2(config-router)#network 172.16.3.0 0.0.0.3 area 0
R2(config-router)#
03:06:00: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.5 on Serial0/0/0 from LOADING to FULL, Loading Done

R2(config-router)#network 192.168.10.8 0.0.0.3 area 0
R2(config-router)#network 209.165.200.224 0.0.0.31 area 0
R2(config-router)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R2#

```

เปิดใช้งาน OSPF process 1 บน R2 และประกาศ network ที่เกี่ยวข้องทั้งหมด ระหว่างตั้งค่าจะเห็นข้อความ ADJCHG แจ้งว่า OSPF adjacency กับ R1 (192.168.10.5) เปลี่ยนสถานะจาก LOADING เป็น FULL แสดงว่าสร้างความสัมพันธ์เพื่อนบ้าน OSPF สำเร็จ จากนั้นบันทึกการตั้งค่าด้วย copy running-config startup-config

```

R3(config)#router ospf 1
R3(config-router)#network 192.168.1.0 0.0.0.255 area 0
R3(config-router)#network 192.168.10.4 0.0.0.3 area 0
R3(config-router)#
00:07:45: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.5 on Serial0/0/0 from LOADING to FULL, Loading Done

R3(config-router)#network 192.168.10.8 0.0.0.3 area 0
R3(config-router)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#
00:07:58: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.9 on Serial0/0/1 from LOADING to FULL, Loading Done

R3#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R3#show ip ospf neighbor

```

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.10.5	0	FULL/ -	00:00:34	192.168.10.5	Serial0/0/0
192.168.10.9	0	FULL/ -	00:00:38	192.168.10.9	Serial0/0/1

```

R3#

```

เปิดใช้งาน OSPF และประกาศ network ทั้งหมดของ R3 พบข้อความ adjacency กับ R1 และ R2 เปลี่ยนเป็น FULL ทั้งคู่ จากนั้นใช้คำสั่ง show ip ospf neighbor ตรวจสอบ ผลลัพธ์ยืนยันว่า R3 มีสถานะ FULL กับทั้ง R1 (192.168.10.5) และ R2 (192.168.10.9) แสดงว่าเครือข่าย OSPF เชื่อมต่อสมบูรณ์ครบทุกลิงก์

Step 1: Test connectivity between all end devices.

Before modifying the OSPF settings, verify that all PCs can ping the web server and each other.

PC1 ping PC2 ,PC1 ping PC3 ,PC1 ping Web Server

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.2.2
Invalid Command.

C:\>ping 172.16.2.2
Pinging 172.16.2.2 with 32 bytes of data:
Request timed out.
Reply from 172.16.2.2: bytes=32 time=5ms TTL=126
Reply from 172.16.2.2: bytes=32 time=16ms TTL=126
Reply from 172.16.2.2: bytes=32 time=1ms TTL=126
Ping statistics for 172.16.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 16ms, Average = 7ms
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>ping 64.100.1.2
Pinging 64.100.1.2 with 32 bytes of data:
Request timed out.
Reply from 64.100.1.2: bytes=32 time=2ms TTL=125
Reply from 64.100.1.2: bytes=32 time=2ms TTL=125
Reply from 64.100.1.2: bytes=32 time=2ms TTL=125
Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms
C:\>
```

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PC2 ping PC1 ,PC2 ping PC3,PC2 ping Web Server

```
PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.1.2
Pinging 172.16.1.2 with 32 bytes of data:
Reply from 172.16.1.2: bytes=32 time=6ms TTL=126
Reply from 172.16.1.2: bytes=32 time=8ms TTL=126
Reply from 172.16.1.2: bytes=32 time=1ms TTL=126
Reply from 172.16.1.2: bytes=32 time=5ms TTL=126
Ping statistics for 172.16.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 6ms, Average = 4ms
C:\>ping 172.16.2.2
Pinging 172.16.2.2 with 32 bytes of data:
Reply from 172.16.2.2: bytes=32 time<1ms TTL=128
Reply from 172.16.2.2: bytes=32 time<1ms TTL=128
Reply from 172.16.2.2: bytes=32 time<1ms TTL=128
Reply from 172.16.2.2: bytes=32 time=4ms TTL=128
Ping statistics for 172.16.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time=5ms TTL=126
Reply from 192.168.1.2: bytes=32 time=6ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 6ms, Average = 3ms
C:\>ping 64.100.1.2
Pinging 64.100.1.2 with 32 bytes of data:
Reply from 64.100.1.2: bytes=32 time=8ms TTL=126
Reply from 64.100.1.2: bytes=32 time=1ms TTL=126
Reply from 64.100.1.2: bytes=32 time=17ms TTL=126
Reply from 64.100.1.2: bytes=32 time=1ms TTL=126
Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 17ms, Average = 6ms
C:\>
```

มีการตอบกลับทุกตัวแสดงว่าถูกต้อง

PC3 ping PC1, PC3 ping PC2, PC3 ping Web Server

```
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.1.2

Pinging 172.16.1.2 with 32 bytes of data:

Reply from 172.16.1.2: bytes=32 time=5ms TTL=126
Reply from 172.16.1.2: bytes=32 time=1ms TTL=126
Reply from 172.16.1.2: bytes=32 time=4ms TTL=126
Reply from 172.16.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 172.16.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 5ms, Average = 2ms

C:\>ping 172.16.2.2

Pinging 172.16.2.2 with 32 bytes of data:

Reply from 172.16.2.2: bytes=32 time=6ms TTL=126
Reply from 172.16.2.2: bytes=32 time=1ms TTL=126
Reply from 172.16.2.2: bytes=32 time=4ms TTL=126
Reply from 172.16.2.2: bytes=32 time=4ms TTL=126

Ping statistics for 172.16.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 6ms, Average = 3ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=6ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time=5ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 3ms

C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 64.100.1.2: bytes=32 time=30ms TTL=125
Reply from 64.100.1.2: bytes=32 time=30ms TTL=125
Reply from 64.100.1.2: bytes=32 time=25ms TTL=125
Reply from 64.100.1.2: bytes=32 time=14ms TTL=125

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 14ms, Maximum = 30ms, Average = 24ms

C:\>
```

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Step 2: Adjust the hello and dead timers between R1 and R2.

- a. Enter the following commands on **R1**.

```
R1(config)# interface s0/0/0
R1(config-if)# ip ospf hello-interval 15
R1(config-if)# ip ospf dead-interval 60
```

```
R1>
R1>ena
R1#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface s0/0/0
R1(config-if)#ip ospf hello-interval 15
R1(config-if)#ip ospf dead-interval 60
R1(config-if)#
```

เข้าสู่อินเทอร์เฟซ S0/0/0 ของ R1 แล้วปรับค่า hello-interval เป็น 15 วินาที และ dead-interval เป็น 60 วินาที ตามที่โจทย์กำหนด เพื่อทดสอบผลกระทบของการตั้งค่า timer ที่ไม่ตรงกันระหว่างสองฝั่ง

- b. After a short period of time, the OSPF connection with **R2** will fail, as shown in the router output.

```
00:02:40: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Dead timer expired
```

```
00:02:40: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
```

Both sides of the connection need to have the same timer values in order for the adjacency to be maintained. Identify the interface on R2 that is connected to R1. Adjust the timers on the R2 interface to match the settings on **R1**.

```
R2(config)# interface s0/0/0
R2(config-if)# ip ospf hello-interval 15
R2(config-if)# ip ospf dead-interval 60
```

After a brief period of time you should see a status message that indicates that the OSPF adjacency has been reestablished.

```
00:21:52: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.5 on Serial0/0/0 from LOADING to FULL, Loading Done
```

```
R2>ena
R2#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#interface s0/0/0
R2(config-if)#
03:56:10: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.5 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Dead timer expired

03:56:10: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.5 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached

R2(config-if)#ip ospf hello-interval 15
R2(config-if)#ip ospf dead-interval 60
R2(config-if)#
03:56:31: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.5 on Serial0/0/0 from LOADING to FULL, Loading Done
```

ก่อนปรับค่า timer บน R2 จะเห็นข้อความแจ้งว่า OSPF adjacency กับ R1 (192.168.10.5) ล่มลง (จาก FULL เป็น DOWN) เนื่องจาก dead timer หมดเวลา เพราะค่า timer ทั้งสองฝั่งไม่ตรงกัน หลังจากปรับ hello-interval เป็น 15 และ dead-interval เป็น 60 ให้ตรงกับ R1 แล้ว adjacency จะกลับมาเป็น FULL อีกครั้งภายในเวลาไม่นาน ยืนยันว่าค่า timer ทั้งสองฝั่งต้องตรงกันเสมอ

Step 3: Adjust the bandwidth setting on R1.

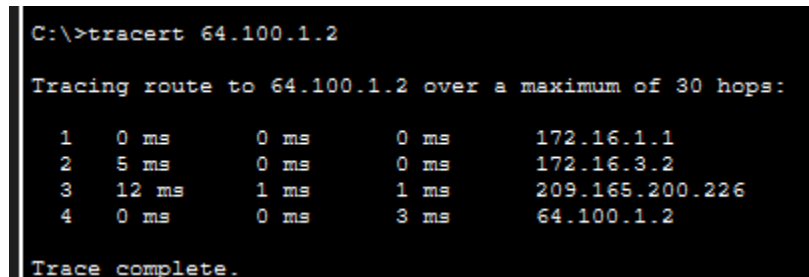
- a. Trace the path between **PC1** and the web server located at 64.100.1.2. Notice that the path from **PC1** to 64.100.1.2 is routed through **R2**. OSPF prefers the lower cost path.

```
C:\> tracert 64.100.1.2
```

Tracing route to 64.100.1.2 over a maximum of 30 hops:

```
 1  1 ms  0 ms  8 ms 172.16.1.1
 2  0 ms  1 ms  0 ms 172.16.3.2
 3  1 ms  9 ms  2 ms 209.165.200.226
 4  *  1 ms  0 ms 64.100.1.2
```

Trace complete.



```
C:\>tracert 64.100.1.2

Tracing route to 64.100.1.2 over a maximum of 30 hops:

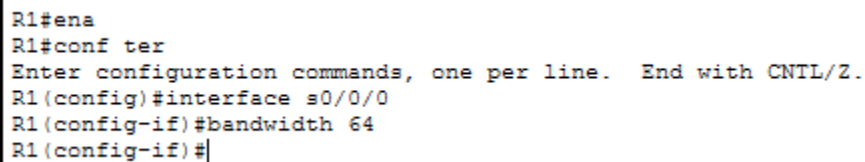
  0  0 ms    0 ms    0 ms    172.16.1.1
  1  5 ms    0 ms    0 ms    172.16.3.2
  2 12 ms    1 ms    1 ms    209.165.200.226
  3  0 ms    0 ms    3 ms    64.100.1.2

Trace complete.
```

ผลลัพธ์ tracert จาก PC1 ไปยัง 64.100.1.2 แสดงเส้นทางผ่าน 172.16.1.1 → 172.16.3.2 (R2) → 209.165.200.226 → 64.100.1.2 คือเส้นทางผ่าน R2 เนื่องจาก OSPF เลือกเส้นทางที่มี cost ต่ำที่สุด

- b. On the **R1** Serial 0/0/0 interface, set the bandwidth to 64 Kb/s. This does not change the actual port speed, only the metric that the OSPF process on **R1** will use to calculate best routes.

```
R1(config-if)# bandwidth 64
```



```
R1#ena
R1#conf ter
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface s0/0/0
R1(config-if)#bandwidth 64
R1(config-if)#
```

- c. Trace the path between **PC1** and the web server located at 64.100.1.2. Notice that the path from **PC1** to 64.100.1.2 is redirected through **R3**. OSPF prefers the lower cost path.

```
C:\> tracert 64.100.1.2
```

Tracing route to 64.100.1.2 over a maximum of 30 hops:

```
 1  1 ms  0 ms  3 ms 172.16.1.1
 2  8 ms  1 ms  1 ms 192.168.10.6
```

```
3 2 ms 0 ms 2 ms 172.16.3.2
4 2 ms 3 ms 1 ms 209.165.200.226
5 2 ms 11 ms 11 ms 64.100.1.2
```

Trace complete.

```
C:\>tracert 64.100.1.2

Tracing route to 64.100.1.2 over a maximum of 30 hops:

  0  0 ms  0 ms  0 ms  172.16.1.1
  1  0 ms  6 ms  1 ms  192.168.10.6
  2  0 ms  0 ms  0 ms  172.16.3.2
  3  1 ms  1 ms  6 ms  209.165.200.226
  4  0 ms  1 ms  1 ms  64.100.1.2

Trace complete.

C:\>
```

หลังปรับค่า bandwidth บนอินเทอร์เฟซ S0/0/0 ของ R1 เป็น 64 Kb/s ผล tracert จาก PC1 เปลี่ยนเส้นทางเป็นผ่าน 172.16.1.1 → 192.168.10.6 (R3) → 172.16.3.2 (R2) → 209.165.200.226 → 64.100.1.2 แสดงให้เห็นว่า OSPF คำนวณ cost ใหม่และเลือกเส้นทางผ่าน R3 แทน เนื่องจากลิงก์ R1-R2 มี cost สูงขึ้นจากการลด bandwidth ลง

Part 2: Verify Connectivity

Verify that all PCs can ping the web server and each other.

PC1 ping PC2 ,PC1 ping PC3 ,PC1 ping Web Server

```
C:\>
C:\>ping 172.16.2.2

Pinging 172.16.2.2 with 32 bytes of data:

Reply from 172.16.2.2: bytes=32 time=9ms TTL=126
Reply from 172.16.2.2: bytes=32 time=17ms TTL=126
Reply from 172.16.2.2: bytes=32 time=1ms TTL=126
Reply from 172.16.2.2: bytes=32 time=17ms TTL=126

Ping statistics for 172.16.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 17ms, Average = 11ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=7ms TTL=126
Reply from 192.168.1.2: bytes=32 time=5ms TTL=126
Reply from 192.168.1.2: bytes=32 time=3ms TTL=126
Reply from 192.168.1.2: bytes=32 time=5ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 3ms, Maximum = 7ms, Average = 5ms

C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 64.100.1.2: bytes=32 time=25ms TTL=125
Reply from 64.100.1.2: bytes=32 time=30ms TTL=125
Reply from 64.100.1.2: bytes=32 time=2ms TTL=125
Reply from 64.100.1.2: bytes=32 time=6ms TTL=125

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 30ms, Average = 15ms

C:\>|
```

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PC2 ping PC1 ,PC2 ping PC3,PC2 ping Web Server

```
C:\>
C:\>ping 172.16.1.2

Pinging 172.16.1.2 with 32 bytes of data:

Reply from 172.16.1.2: bytes=32 time=10ms TTL=125
Reply from 172.16.1.2: bytes=32 time=6ms TTL=125
Reply from 172.16.1.2: bytes=32 time=5ms TTL=125
Reply from 172.16.1.2: bytes=32 time=7ms TTL=125

Ping statistics for 172.16.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 10ms, Average = 7ms

C:\>ping 172.16.2.2

Pinging 172.16.2.2 with 32 bytes of data:

Reply from 172.16.2.2: bytes=32 time=2ms TTL=128
Reply from 172.16.2.2: bytes=32 time=4ms TTL=128
Reply from 172.16.2.2: bytes=32 time=3ms TTL=128
Reply from 172.16.2.2: bytes=32 time=3ms TTL=128

Ping statistics for 172.16.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 3ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=8ms TTL=126
Reply from 192.168.1.2: bytes=32 time=5ms TTL=126
Reply from 192.168.1.2: bytes=32 time=5ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 8ms, Average = 4ms

C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 64.100.1.2: bytes=32 time=1ms TTL=126
Reply from 64.100.1.2: bytes=32 time=12ms TTL=126
Reply from 64.100.1.2: bytes=32 time=20ms TTL=126
Reply from 64.100.1.2: bytes=32 time=3ms TTL=126

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 20ms, Average = 9ms

C:\>
```

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PC3 ping PC1, PC3 ping PC2, PC3 ping Web Server

```
C:\>ping 172.16.1.2

Pinging 172.16.1.2 with 32 bytes of data:

Reply from 172.16.1.2: bytes=32 time=1ms TTL=126
Reply from 172.16.1.2: bytes=32 time=5ms TTL=126
Reply from 172.16.1.2: bytes=32 time=1ms TTL=126
Reply from 172.16.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 172.16.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 5ms, Average = 2ms

C:\>ping 172.16.2.2

Pinging 172.16.2.2 with 32 bytes of data:

Reply from 172.16.2.2: bytes=32 time=8ms TTL=126
Reply from 172.16.2.2: bytes=32 time=1ms TTL=126
Reply from 172.16.2.2: bytes=32 time=1ms TTL=126
Reply from 172.16.2.2: bytes=32 time=1ms TTL=126

Ping statistics for 172.16.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 8ms, Average = 2ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=2ms TTL=128
Reply from 192.168.1.2: bytes=32 time=3ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 2ms

C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 64.100.1.2: bytes=32 time=39ms TTL=125
Reply from 64.100.1.2: bytes=32 time=51ms TTL=125
Reply from 64.100.1.2: bytes=32 time=2ms TTL=125
Reply from 64.100.1.2: bytes=32 time=18ms TTL=125

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 51ms, Average = 27ms

C:\>|
```

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